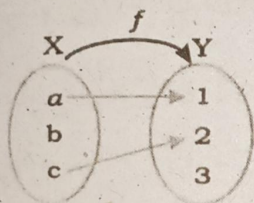


CHAPTER # 08 FUNCTIONS & GRAPHS

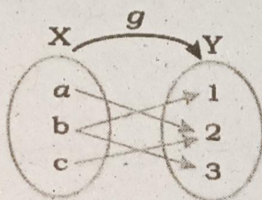
EXERCISE 8.1

Q.1 Let $X = \{a, b, c\}$ and $Y = \{1, 2, 3\}$. State whether the relations indicated by the following figures are functions from X to Y .

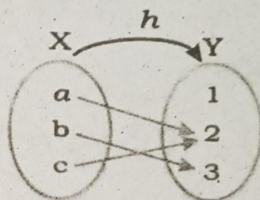
(i)



(ii)



(iii)



SOLUTION:

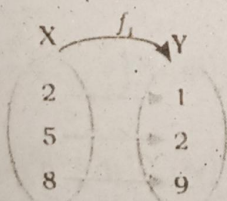
(i) not a function

(ii) not a function

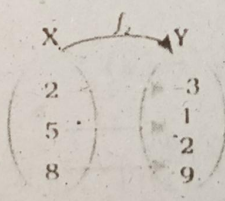
(iii) a function

Q.2 Which of the following functions are injective, surjective and bijective. Give reason.

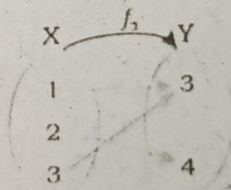
(i)



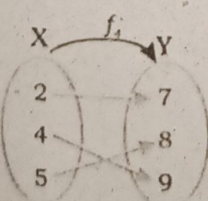
(ii)



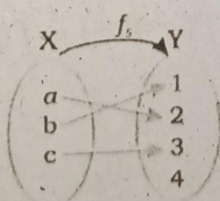
(iii)



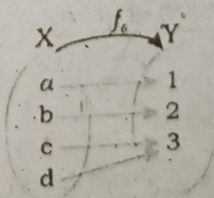
(iv)



(v)



(vi)



SOLUTION:

(i) Bijective because f_1 is one to one and onto

(ii) injective because distinct elements of domain have distinct images

(iii) a surjective because range of $f_3 = Y$

A Help Book of Mathematics XI

(iv) Bijective because f_4 is one to one and onto

(v) injective because distinct elements of domain have distinct images

(vi) surjective because range of $f_6 = Y$

Q.3 If function $f: Z \rightarrow R$ defined by $f(x) = x^3 + 1$, find the following if possible:

(i) $f(0)$ (ii) $f(5)$ (iii) $f\left(\frac{3}{4}\right)$ (iv) $f(-2)$

SOLUTION:

(i) $f(0) = (0)^3 + 1 = 0 + 1 = 1$

(ii) $f(5) = (5)^3 + 1 = 125 + 1 = 126$

(iii) $f\left(\frac{3}{4}\right)$

$\frac{3}{4} \notin Z \Rightarrow f\left(\frac{3}{4}\right)$ is undefined

(iv) $f(-2) = (-2)^3 + 1 = -8 + 1 = -7$

Q.4 If $f(x) = x^3$, find the values of:

(i) $f(2)$ (ii) $f(-10)$ (iii) $f\left(\frac{1}{2}\right)$

(iv) $f(5a)$ (v) $f\left(\frac{a}{3}\right)$ (vi) $f(a+h)$

(vii) $\frac{f(a+h)-f(a-h)}{2h}$ (viii) $f(a+h) - f(a-h)$

SOLUTION:

(i) $f(2) = (2)^3 = 8$

(ii) $f(-10) = (-10)^3 = -1000$

(iii) $f\left(\frac{1}{2}\right) = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$

(iv) $f(5a) = (5a)^3 = 125a^3$

$$(v) f\left(\frac{a}{3}\right) = \left(\frac{a}{3}\right)^3 = \frac{a^3}{27}$$

$$(vi) f(a+h) = (a+h)^3 = a^3 + 3a^2h + 3ah^2 + h^3$$

$$(vii) \frac{f(a+h)-f(a-h)}{2h}$$

$$f(a-h) = (a-h)^3 = a^3 - 3a^2h + 3ah^2 - h^3$$

$$f(a+h) - f(a-h) = a^3 + 3a^2h + 3ah^2 + h^3$$

$$-(a^3 - 3a^2h + 3ah^2 - h^3)$$

$$f(a+h) - f(a-h) = 6a^2h + 2h^3$$

$$f(a+h) - f(a-h) = \frac{6a^2h + 2h^3}{2h} = \frac{2h(3a^2 + h^2)}{2h} = 3a^2 + h^2$$

$$(viii) f(a+h) - f(a-h)$$

$$f(a-h) = (a-h)^3 = a^3 - 3a^2h + 3ah^2 - h^3$$

$$f(a+h) - f(a-h) = 6a^2h + 2h^3$$

Q.5 Given that $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^2$, find:

$$(i) \frac{f(5+h)-f(5)}{h}, (h \neq 0) \quad (ii) \frac{f(a+h)-f(a)}{h}, (h \neq 0)$$

SOLUTION:

$$f(x) = x^2$$

$$(i) \frac{f(5+h)-f(5)}{h}$$

$$f(5+h) = (5+h)^2 = 25 + 10h + h^2$$

$$f(5) = (5)^2 = 25$$

$$f(5+h) - f(5) = 10h + h^2$$

$$\frac{f(5+h) - f(5)}{h} = \frac{10h + h^2}{h} = \frac{h(10 + h)}{h} = 10 + h$$

$$(ii) \frac{f(a+h)-f(a)}{h}$$

$$f(a+h) = (a+h)^2 = a^2 + 2ah + h^2$$

$$f(a) = (a)^2 = a^2$$

$$f(a+h) - f(a) = 2ah + h^2$$

$$\frac{f(a+h)-f(a)}{h} = \frac{2ah+h^2}{h} = \frac{h(2a+h)}{h} = 2a + h$$

Q.6 The domain of the function $f(x) = 5x + 1$ is $\{0, 1, 2, 3, 4, 5\}$. Find its range.

SOLUTION:

$$f(x) = 5x + 1$$

$$\text{Domain} = \{0, 1, 2, 3, 4, 5\}$$

$$f(0) = 5(0) + 1 = 1$$

$$f(1) = 5(1) + 1 = 6$$

$$f(2) = 5(2) + 1 = 11$$

$$f(3) = 5(3) + 1 = 16$$

$$f(4) = 5(4) + 1 = 21$$

$$f(5) = 5(5) + 1 = 26$$

$$\text{Range} = \{1, 6, 11, 16, 21, 26\}$$

Q.7 Find the domain of the following functions:

$$(i) f(x) = x^2 + 3$$

$$(ii) f(x) = \sqrt{x+3}$$

$$(iii) f(x) = \frac{1}{x}$$

$$(iv) f(x) = \sqrt{x-5}$$

$$(v) f(x) = \frac{x+1}{x-2}$$

$$(vi) f(x) = \frac{x^2}{x^2-9}$$

SOLUTION:

$$(i) f(x) = x^2 + 3$$

$$\text{Domain} = \mathbb{R}$$

$$(ii) f(x) = \sqrt{x+3}$$

$$x+3 \geq 0$$

$$x \geq -3 \quad \forall x \in \mathbb{R}$$

$$\text{Domain} = [-3, \infty)$$

$$(iii) f(x) = \frac{1}{x}$$

$$x \neq 0$$

$$\text{Domain} = \mathbb{R} - \{0\}$$

$$(iv) f(x) = \sqrt{x-5}$$

$$x-5 \geq 0$$

$$x \geq 5$$

$$\text{Domain} = [5, \infty)$$

$$(v) f(x) = \frac{x+1}{x-2}$$

$$x-2 \neq 0$$

$$x \neq 2$$

$$\text{Domain} = \mathbb{R} - \{2\}$$

$$(vi) f(x) = \frac{x^2}{x^2-9}$$

$$x^2-9 \neq 0$$

$$x^2 \neq 9$$

$$x \neq \pm 3$$

$$\text{Domain} = \mathbb{R} - \{\pm 3\}$$

Q.8 The domain of the function $f(x)$ is $\{1, 2, 3, 4, 5\}$. Find the range of:

$$f(x) = 5x^2 + 3$$

$$(ii) f(x) = \frac{x}{x+1}$$

SOLUTION:

$$f(x) = 5x^2 + 3$$

$$\text{Domain} = \{1, 2, 3, 4, 5\}$$

$$f(1) = 5(1)^2 + 3 = 8$$

$$f(2) = 5(2)^2 + 3 = 23$$

$$f(3) = 5(3)^2 + 3 = 48$$

$$f(4) = 5(4)^2 + 3 = 83$$

$$f(5) = 5(5)^2 + 3 = 128$$

$$(ii) f(x) = \frac{x}{x+1}$$

$$\text{Domain} = \{1, 2, 3, 4, 5\}$$

$$f(1) = \frac{1}{1+1} = \frac{1}{2}$$

$$f(2) = \frac{2}{2+1} = \frac{2}{3}$$

$$f(3) = \frac{3}{3+1} = \frac{3}{4}$$

$$f(4) = \frac{4}{4+1} = \frac{4}{5}$$

$$f(5) = \frac{5}{5+1} = \frac{5}{6}$$

Q.9 Find the range of the function $f(x) = \sqrt{x^2 + 1}$.

SOLUTION:

$x^2 + 1$ can never be negative then Domain = R

Q.10 Find the domain and range of the function $f(x) = \frac{1}{1+x^2}$.

SOLUTION:

$$D(f) = R$$

$$\text{Range} = y > 0 \forall y \in R$$

$x^2 + 1$ can never be zero then Domain = $\mathbb{R}$

Q.11 Find the domain and range of $f(x) = \frac{1}{\sqrt{25-x}}$.

SOLUTION:

$$25 - x > 0$$

$$25 > x$$

$$x < 25$$

$$\text{Domain} = (-\infty, 25)$$

$$\mathbb{R}(f) = \mathbb{R}^+$$

Q.12 A function is defined by $f(x) = ax + b$. The images of 1 and 5 are -2 and 10 respectively.

(i) Find the value of a and b .

(ii) The domain of $f = \{1, 3, 5\}$, find the range of f .

SOLUTION:

(i) Find the value of a and b .

$$f(x) = ax + b$$

$$f(1) = -2$$

$$a(1) + b = -2$$

$$a + b = -2 \rightarrow (1)$$

$$f(5) = 10$$

$$a(5) + b = 10$$

$$5a + b = 10 \rightarrow (2)$$

$$\text{By } (2) - (1)$$

$$4a = 10 + 2$$

$$4a = 12$$

$$\boxed{a = 3}$$

$$(1) \Rightarrow b = -2 - a$$

$$b = -2 - 3$$

$$\boxed{b = -5}$$

(ii) The domain of $f = \{1, 3, 5\}$, find the range of f .

$$f(x) = ax + b$$

$$f(x) = 3x - 5$$

$$f(1) = 3(1) - 5 = -2$$

$$f(3) = 3(3) - 5 = 4$$

$$f(5) = 3(5) - 5 = 10$$

$$\text{Range} = \{-2, 4, 10\}$$

Q.13 If $f(x) = x^3 - ax^2 + bx + 1$, find the values of a and b , where $f(2) = -3$ and $f(-1) = 0$.

SOLUTION:

$$f(x) = x^3 - ax^2 + bx + 1$$

$$f(-1) = (-1)^3 - a(-1)^2 + b(-1) + 1$$

$$0 = -1 - a(1) - b + 1$$

$$0 = -1 - a - b + 1$$

$$a + b = 0$$

$$a = -b \rightarrow (1)$$

$$f(2) = (2)^3 - a(2)^2 + b(2) + 1$$

$$-3 = 8 - 4a + 2b + 1$$

$$-3 - 9 = -4a + 2b$$

$$-4a + 2b = -12$$

$$-2(-b) + b = -6$$

$$2b + b = -6$$

$$3b = -6$$

$$\boxed{b = -2}$$

$$(1) \Rightarrow a = -(-2)$$

$$\boxed{a = 2}$$

EXERCISE 8.2

Q.1 If $f(x) = -5x + 1$, find the values of:

(i) $f^{-1}(36)$ (ii) $f^{-1}(-14)$ (iii) $f^{-1}(0)$ (iv) $f^{-1}(a)$

SOLUTION:

$$f(x) = -5x + 1$$

(i) $f^{-1}(36) = c$

$$36 = f(c)$$

$$36 = -5c + 1$$

$$5c = 1 - 36$$

$$5c = -35$$

$$\boxed{c = -7}$$

$$f^{-1}(36) = -7$$

(ii) $f^{-1}(-14) = c$

$$-14 = f(c)$$

$$-14 = -5c + 1$$

$$5c = 1 + 14$$

$$5c = 15$$

$$\boxed{c = 3}$$

$$f^{-1}(-14) = 3$$

(iii) $f^{-1}(0) = c$

$$0 = f(c)$$

$$0 = -5c + 1$$

$$5c = 1$$

$$\boxed{c = \frac{1}{5}}$$

$$f^{-1}(0) = \frac{1}{5}$$

$$(iv) f^{-1}(a) = c$$

$$a = -5c + 1$$

$$5c = -a + 1$$

$$\boxed{c = \frac{-a+1}{5}}$$

$$f^{-1}(a) = \frac{-a+1}{5}$$

Q.2 Given that $g(t) = \frac{1}{t-5}$, ($t \neq 5$), find the values of:

$$(i) g^{-1}\left(\frac{1}{2}\right)$$

$$(ii) g^{-1}(2)$$

$$(iii) g^{-1}(-1)$$

$$(iv) g^{-1}(a)$$

SOLUTION:

$$g(t) = \frac{1}{t-5}$$

$$(i) g^{-1}\left(\frac{1}{2}\right) = c$$

$$\frac{1}{2} = g(c)$$

$$\frac{1}{2} = \frac{1}{c-5}$$

$$c - 5 = 2$$

$$c = 5 + 2$$

$$\boxed{c = 7}$$

$$g^{-1}\left(\frac{1}{2}\right) = 7$$

$$(ii) g^{-1}(2) = c$$

$$2 = g(c)$$

$$2 = \frac{1}{c-5}$$

$$2(c-5) = 1$$

$$2c - 10 = 1$$

$$2c = 1 + 10$$

$$2c = 11$$

$$\boxed{c = \frac{11}{2}}$$

$$g^{-1}(2) = \frac{11}{2}$$

$$(iii) g^{-1}(-1) = c$$

$$-1 = g(c)$$

$$-1 = \frac{1}{c-5}$$

$$-c + 5 = 1$$

$$5 - 1 = c$$

$$\boxed{c = 4}$$

$$g^{-1}(-1) = 4$$

$$(iv) g^{-1}(a) = c$$

$$a = g(c)$$

$$a = \frac{1}{c-5}$$

$$a(c-5) = 1$$

$$ac - 5a = 1$$

$$ac = 1 + 5a$$

$$\boxed{c = \frac{1+5a}{a}}$$

$$g^{-1}(a) = \frac{1+5a}{a}$$

Q.3 Find the inverse of the following functions:

$$(i) f(x) = 12 - \frac{1}{2}x \quad (ii) f(x) = \frac{1}{2}(x - 3) \quad (iii) f(x) = \frac{2x+1}{5}$$

$$(iv) f(x) = \frac{5}{9}(x - 32) \quad (v) f(x) = 180(x - 2) \quad (vi) h(x) = 2\pi x$$

$$(vii) f(t) = t^2 + 5(t > 0) \quad (viii) f(x) = 5\sqrt{t}, (t \geq 0)$$

$$(ix) f(t) = (t - 5)^3 \quad (x) f(t) = \sqrt[3]{t+1}$$

$$(xi) g(x) = \frac{1}{x-3}, (x \neq 3) \quad (xii) f(x) = \frac{1}{2x+1}, \left(x \neq -\frac{1}{2}\right)$$

SOLUTION:

$$(i) f(x) = 12 - \frac{1}{2}x$$

$$y = 12 - \frac{1}{2}x$$

$$\frac{1}{2}x = 12 - y$$

$$x = 24 - 2y$$

$$\boxed{f^{-1}(x) = 24 - 2x}$$

$$(ii) f(x) = \frac{1}{2}(x - 3)$$

$$y = \frac{1}{2}(x - 3)$$

$$2y = x - 3$$

$$x - 3 = 2y$$

$$x = 3 + 2y$$

$$\boxed{f^{-1}(x) = 3 + 2x}$$

$$(iii) f(x) = \frac{2x+1}{5}$$

$$y = \frac{2x+1}{5}$$

$$5y = 2x + 1$$

$$5y - 1 = 2x$$

$$2x = 5y - 1$$

$$x = \frac{5y-1}{2}$$

$$\boxed{f^{-1}(x) = \frac{5x-1}{2}}$$

$$(iv) f(x) = \frac{5}{9}(x - 32)$$

$$y = \frac{5}{9}(x - 32)$$

$$9y = 5(x - 32)$$

$$9y = 5x - 160$$

$$9y + 160 = 5x$$

$$5x = 9y + 160$$

$$x = \frac{9y+160}{5}$$

$$f^{-1}(x) = \frac{9x + 160}{5}$$

$$\boxed{f^{-1}(x) = 32 + \frac{9x}{5}}$$

$$(v) f(x) = 180(x - 2)$$

$$y = 180(x - 2)$$

$$y = 180x - 360$$

$$y + 360 = 180x$$

$$180x = y + 360$$

$$x = \frac{y+360}{180}$$

$$f^{-1}(x) = \frac{x + 360}{180}$$

$$\boxed{f^{-1}(x) = 2 + \frac{x}{180}}$$

(vi) $h(x) = 2\pi x$

$$y = 2\pi x$$

$$2\pi x = y$$

$$x = \frac{y}{2\pi}$$

$$\boxed{f^{-1}(x) = \frac{x}{2\pi}}$$

(vii) $f(t) = t^2 + 5 (t > 0)$

$$y = t^2 + 5$$

$$y - 5 = t^2$$

$$t^2 = y - 5$$

$$t = \sqrt{y - 5}$$

$t > 0$ then it will be a positive perfect square

$$\boxed{f^{-1}(t) = \sqrt{t - 5}}$$

(viii) $f(x) = 5\sqrt{t}, (t \geq 0)$

$$y = 5\sqrt{t}$$

$$5\sqrt{t} = y$$

$$\sqrt{t} = \frac{y}{5}$$

$$t = \frac{y^2}{25}$$

$$f^{-1}(t) = \frac{t^2}{25}$$

$$(ix) f(t) = (t - 5)^3$$

$$y = (t - 5)^3$$

Taking cube root both sides

$$y^{\frac{1}{3}} = (t - 5)$$

$$t - 5 = y^{\frac{1}{3}}$$

$$t = y^{\frac{1}{3}} + 5$$

$$f^{-1}(t) = t^{\frac{1}{3}} + 5$$

$$(x) f(t) = \sqrt[3]{t + 1}$$

$$y = (t + 1)^{\frac{1}{3}}$$

Cubing both sides

$$y^3 = t + 1$$

$$t + 1 = y^3$$

$$t = y^3 - 1$$

$$f^{-1}(t) = t^3 - 1$$

$$(xi) g(x) = \frac{1}{x-3}, (x \neq 3)$$

$$y = \frac{1}{x-3}$$

$$y(x - 3) = 1$$

$$xy - 3y = 1$$

$$xy = 1 + 3y$$

$$x = \frac{1+3y}{y}$$

$$\boxed{f^{-1}(x) = \frac{1+3x}{x}}$$

$$(xi) f(x) = \frac{1}{2x+1}, \left(x \neq -\frac{1}{2}\right)$$

$$y = \frac{1}{2x+1}$$

$$y(2x+1) = 1$$

$$2xy + y = 1$$

$$2xy = 1 - y$$

$$x = \frac{1-y}{2y}$$

$$\boxed{f^{-1}(x) = \frac{1-x}{2x}}$$

Q.4 Find inverse of f and determine the domain and range of f^{-1} for the real valued function defined by:

$$(i) f(x) = \frac{x-1}{x-3}, x \neq 3$$

$$(ii) f(x) = 5x + 7$$

$$(iii) f(x) = \frac{1}{x+5}, x \neq -5$$

$$(iv) f(x) = (x-5)^2, x \geq 5$$

$$(v) f(x) = \sqrt{x+7}, x \geq -7$$

$$(vi) h(x) = \frac{x-1}{x-4}, x \neq 4$$

SOLUTION:

$$(i) f(x) = \frac{x-1}{x-3}, x \neq 3$$

$$y = \frac{x-1}{x-3}$$

$$y(x-3) = x-1$$

$$xy - 3y = x - 1$$

$$xy - x = 3y - 1$$

$$x(y-1) = 3y-1$$

$$x = \frac{3y-1}{y-1}$$

$$y \neq 1$$

$$\boxed{f^{-1}(x) = \frac{3x-1}{x-1}}$$

$$\text{Range of } f^{-1} = \text{Domain } f = \mathbb{R} - \{3\}$$

$$\text{Domain of } f^{-1} = \text{Range of } f = \mathbb{R} - \{1\}$$

$$(ii) f(x) = 5x + 7$$

$$y = 5x + 7$$

$$y - 7 = 5x$$

$$5x = y - 7$$

$$x = \frac{y-7}{5}$$

$$\boxed{f^{-1}(x) = \frac{x-7}{5}}$$

$$\text{Range of } f^{-1} = \text{Domain } f = \mathbb{R}$$

$$\text{Domain of } f^{-1} = \text{Range of } f = \mathbb{R}$$

$$(iii) f(x) = \frac{1}{x+5}, x \neq -5$$

$$y = \frac{1}{x+5}$$

$$y(x+5) = 1$$

$$xy + 5y = 1$$

$$xy = 1 - 5y$$

$$x = \frac{1-5y}{y}$$

$$y \neq 0$$

$$f^{-1}(x) = \frac{1-5x}{x}$$

Range of $f^{-1} = \text{Domain } f = \mathbb{R} - \{-5\}$

Domain of $f^{-1} = \text{Range of } f = \mathbb{R} - \{0\}$

(iv) $f(x) = (x-5)^2, x \geq 5$

$$y = (x-5)^2$$

Taking positive square root

$$\sqrt{y} = (x-5)$$

$$x-5 = \sqrt{y}$$

$$x = \sqrt{y} + 5$$

$$y \geq 0$$

$$f^{-1}(x) = \sqrt{x} + 5$$

Range of $f^{-1} = \text{Domain } f = [5, \infty)$

Domain of $f^{-1} = \text{Range of } f = [0, \infty)$

(v) $f(x) = \sqrt{x+7}, x \geq -7$

$$y = \sqrt{x+7}$$

Squaring both sides

$$y^2 = x+7$$

$$x+7 = y^2$$

$$x = y^2 - 7$$

$$f^{-1}(x) = x^2 - 7$$

Range of $f^{-1} = \text{Domain } f = [0, \infty) y/y \in \mathbb{R}, y \geq -7$

Domain of $f^{-1} = \text{Range of } f = [0, \infty) x \in \mathbb{R}$

(vi) $h(x) = \frac{x-1}{x-4}, x \neq 4$

$$y = \frac{x-1}{x-4}$$

$$y(x-4) = x-1$$

$$xy - 4y = x - 1$$

$$xy - x = 4y - 1$$

$$x(y-1) = 4y-1$$

$$x = \frac{4y-1}{y-1}$$

$$y \neq 1$$

$$f^{-1}(x) = \frac{4x-1}{x-1}$$

Range of $f^{-1} = \text{Domain } f = \mathbb{R} - \{4\}$

Domain of $f^{-1} = \text{Range of } f = \mathbb{R} - \{1\}$

EXERCISE 8.3

Q.1 Sketch the graph of the following functions:

(i) $y = 5x + 2$

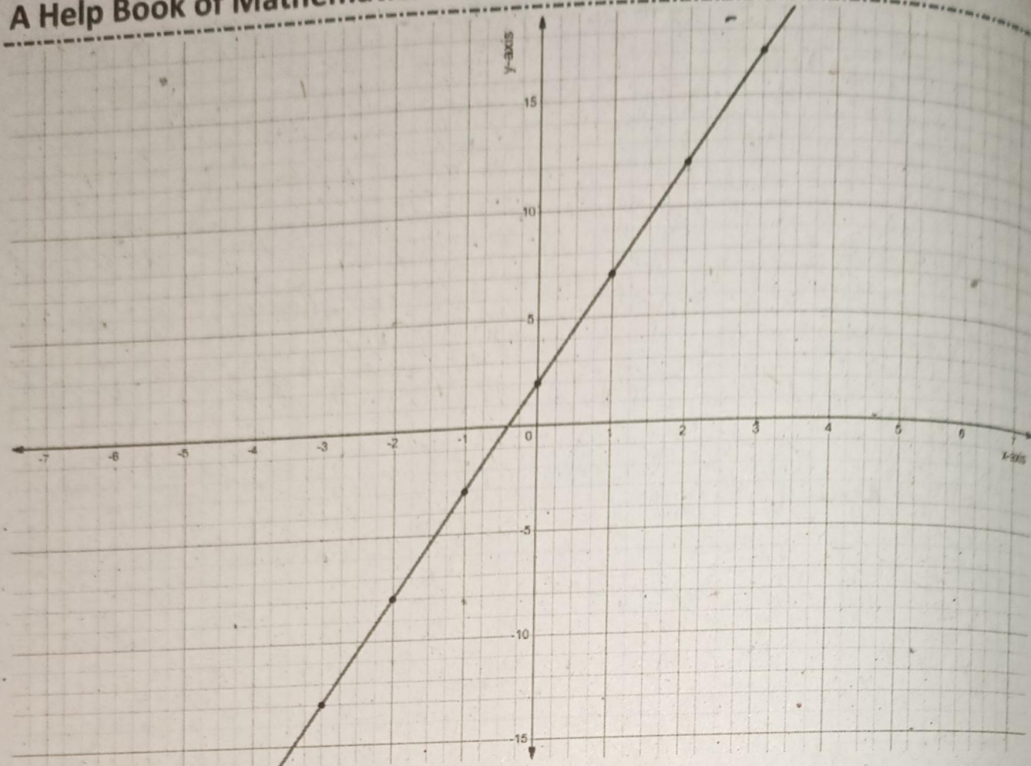
(ii) $y = x^2 + 5$

(iii) $y = x^5$

SOLUTION:

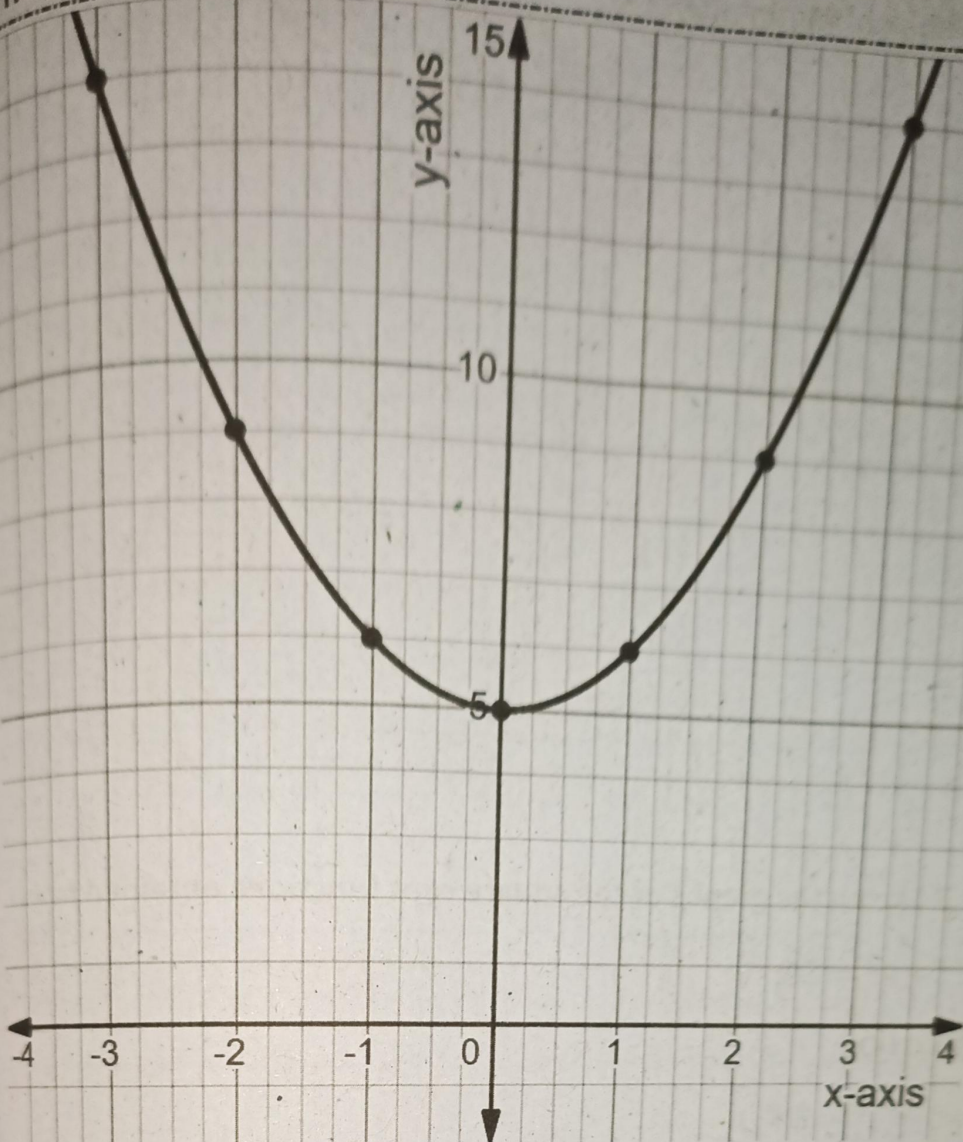
(i) $y = 5x + 2$

x	-3	-2	-1	0	1	2	3
$f(x)$	-13	-8	-3	2	7	12	17



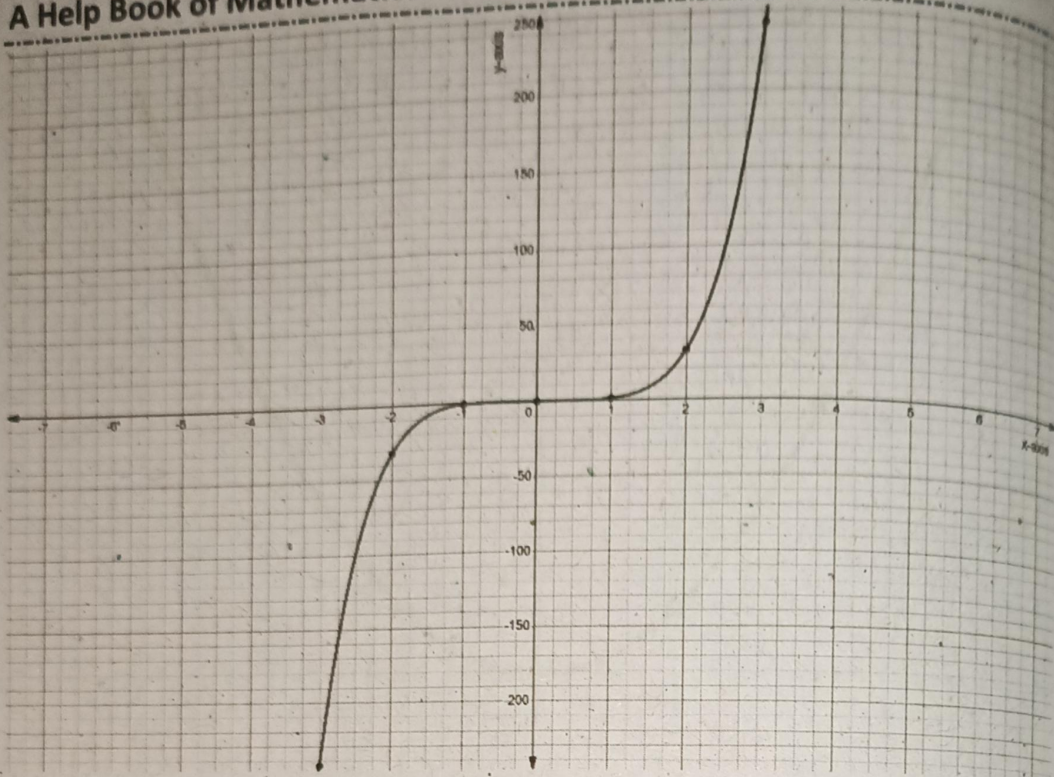
(ii) $y = x^2 + 5$

x	-3	-2	-1	0	1	2	3
$f(x)$	14	9	6	5	6	9	14



(iii) $y = x^5$

x	-3	-2	-1	0	1	2	3
$f(x)$	-243	-32	-1	0	1	32	243



Q.2 Sketch the graphs of the square root functions obtained from

$$x^2 + y^2 = 25.$$

SOLUTION:

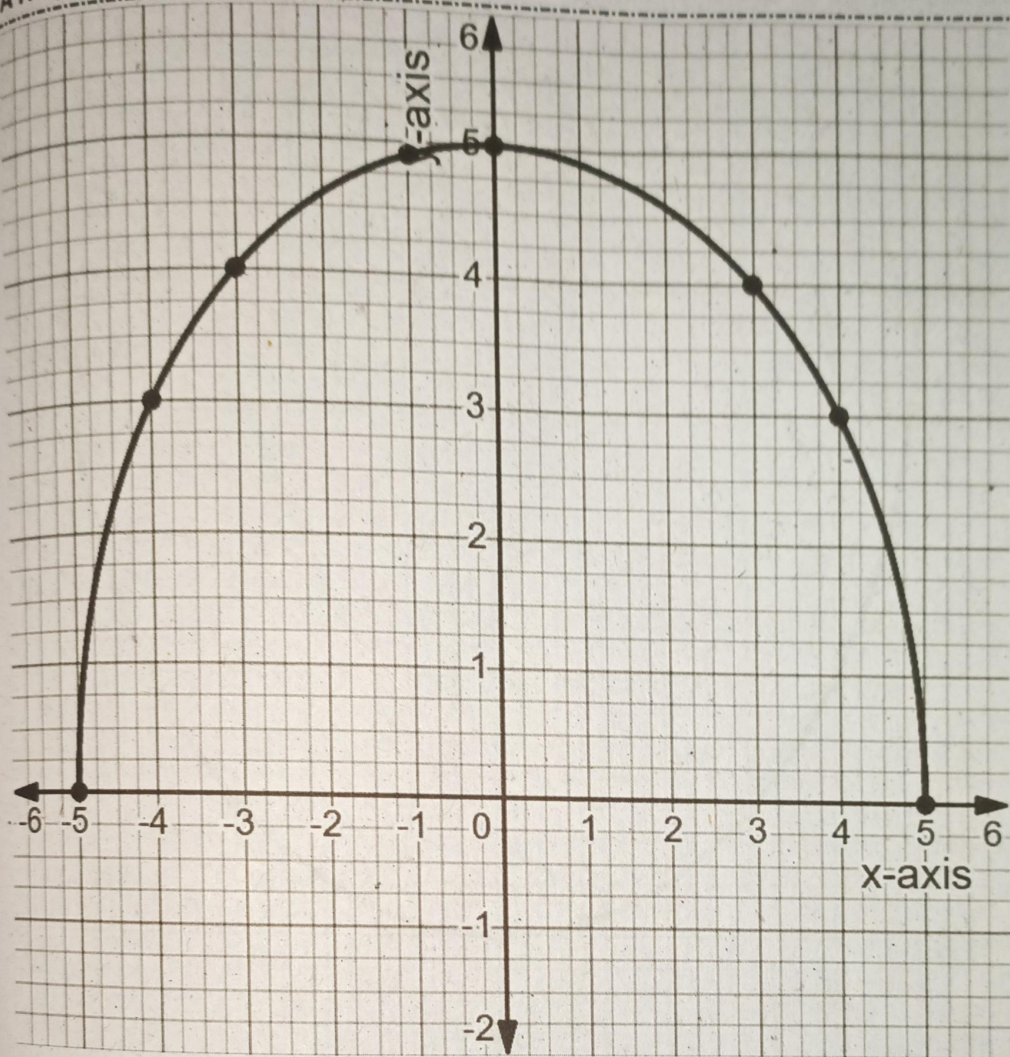
$$x^2 + y^2 = 25$$

$$y^2 = 25 - x^2$$

$$y = \pm \sqrt{25 - x^2}$$

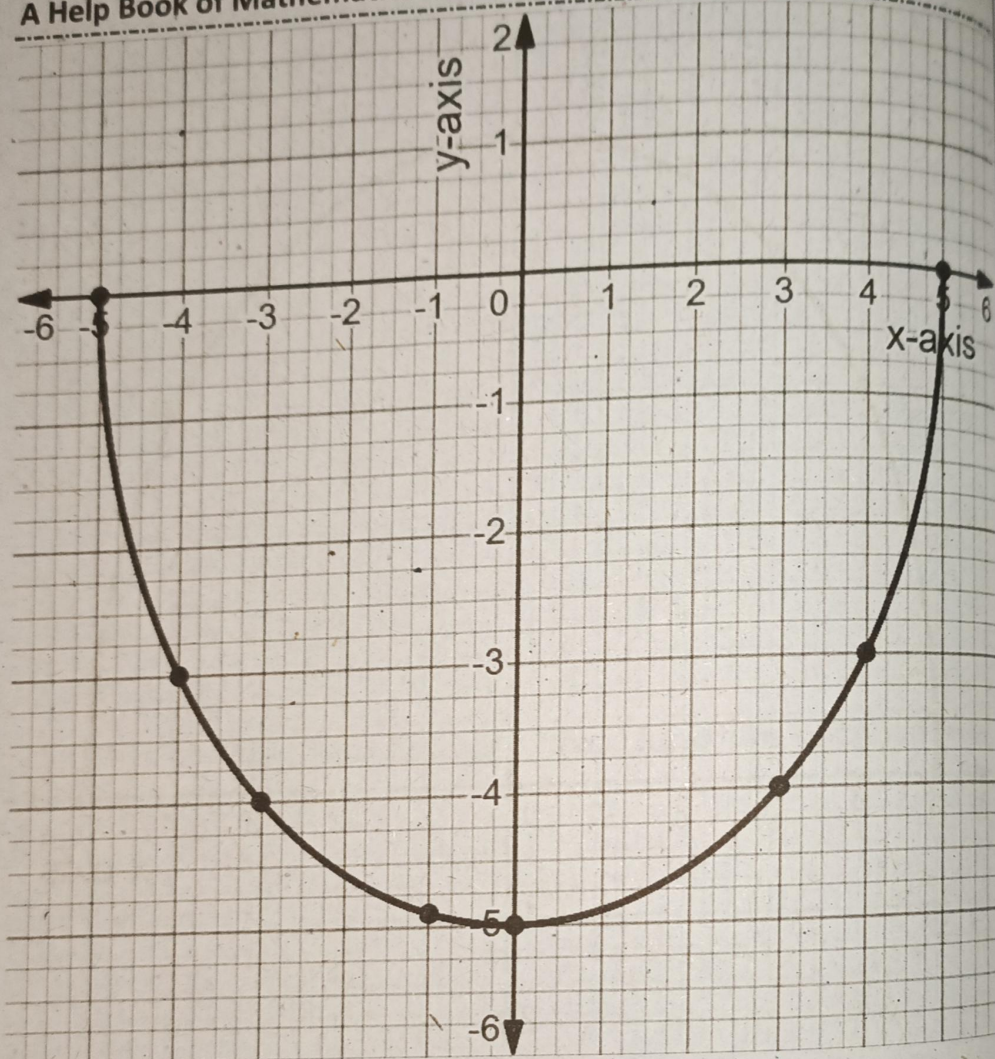
$$y = \sqrt{25 - x^2}$$

x	-5	-4	-3	0	3	4	5
$f(x)$	0	3	4	5	4	3	0



$$y = -\sqrt{25 - x^2}$$

x	-5	-4	-3	0	3	4	5
$f(x)$	0	-3	-4	-5	-4	-3	0



Q.3 Sketch the graph of the following quadratic functions:

(i) $y = x^2 - 2x + 1$

(ii) $y = -3x^2 + 6x$

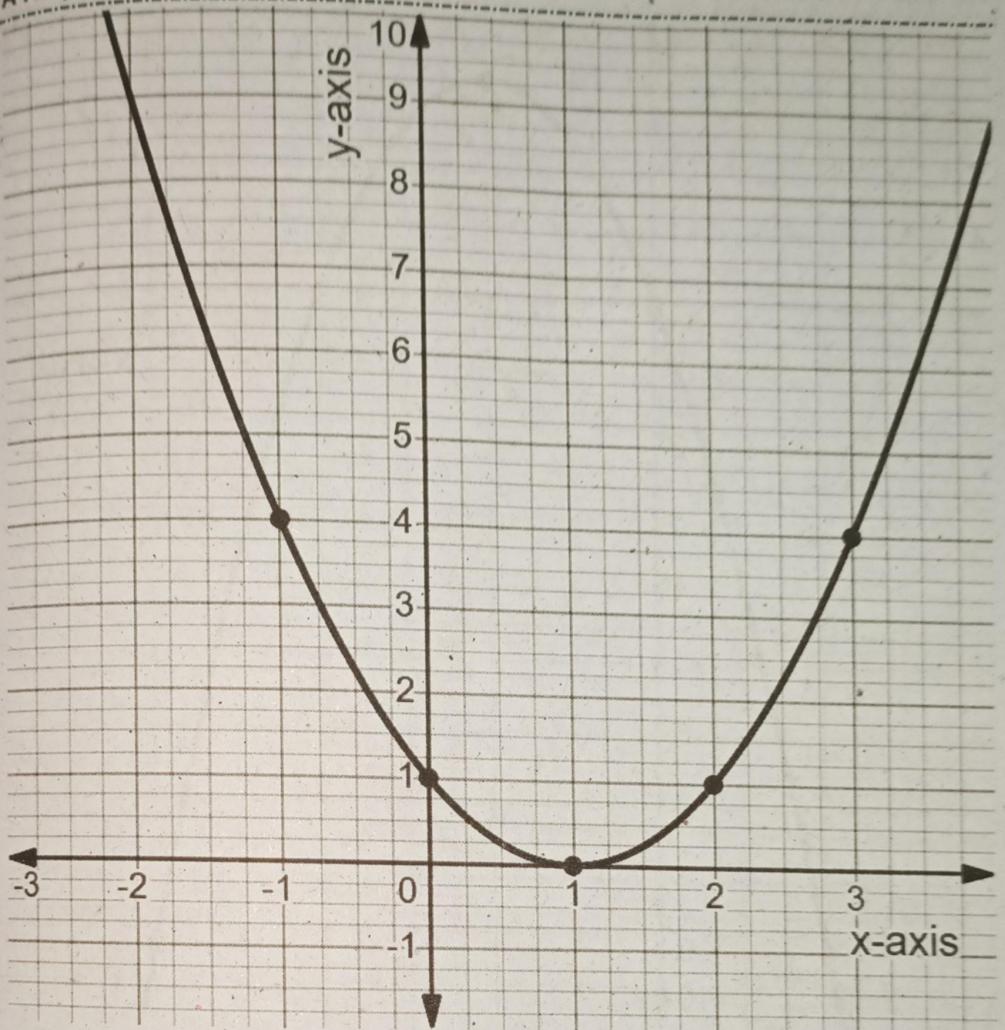
SOLUTION:

(i) $y = x^2 - 2x + 1$

$$y = (x)^2 - 2(x)(1) + (1)^2$$

$$y = (x - 1)^2$$

x	-2	-1	0	1	2	3
$f(x)$	9	4	1	0	1	4



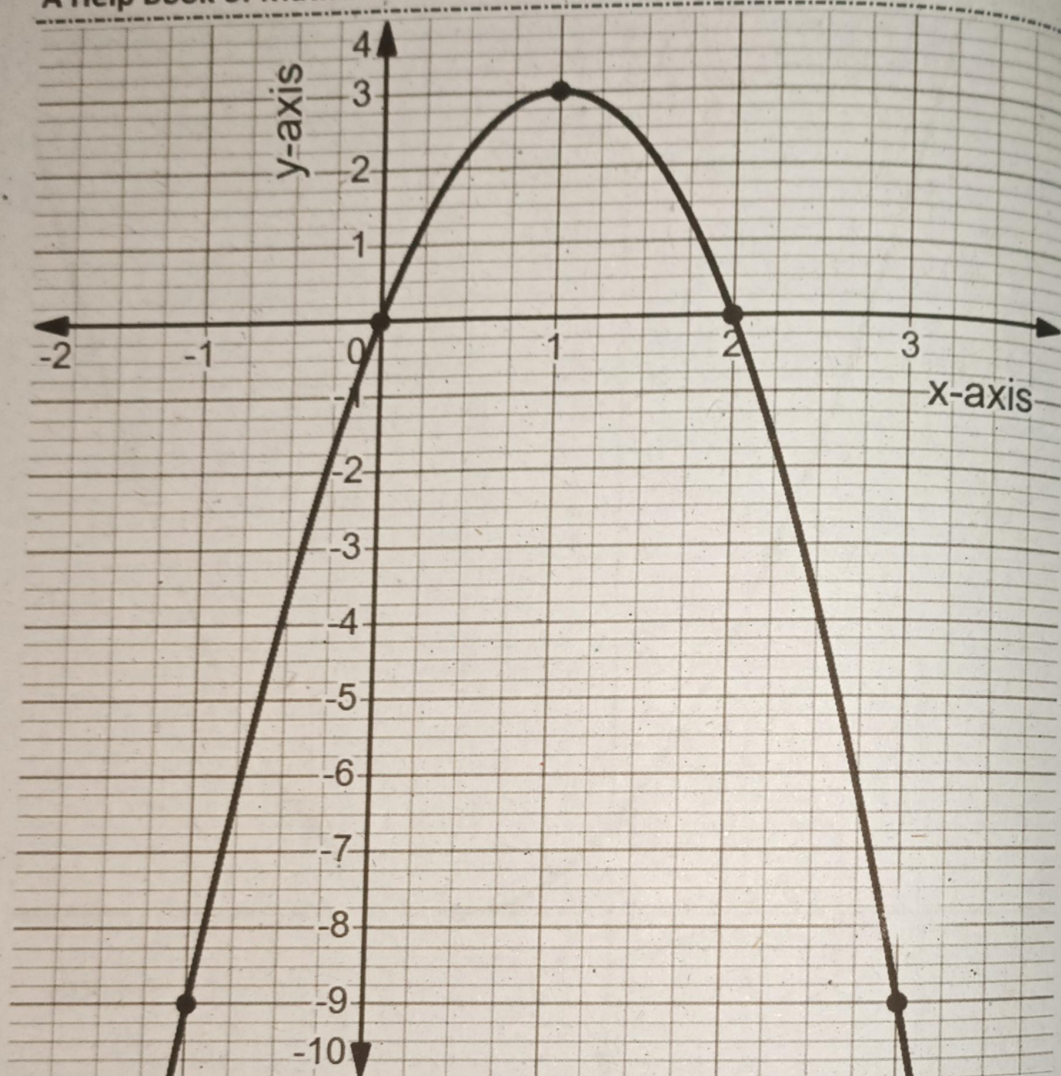
(ii) $y = -3x^2 + 6x$

$$y = -3(x^2 - 2x)$$

$$y = -3(x^2 - 2x + 1 - 1)$$

$$y = -3\{(x - 1)^2 - 1\}$$

$$y = -3(x - 1)^2 + 3$$



Q.4 Sketch the graph of the following functions using factors:

(i) $y = x^2 + 3x + 2$

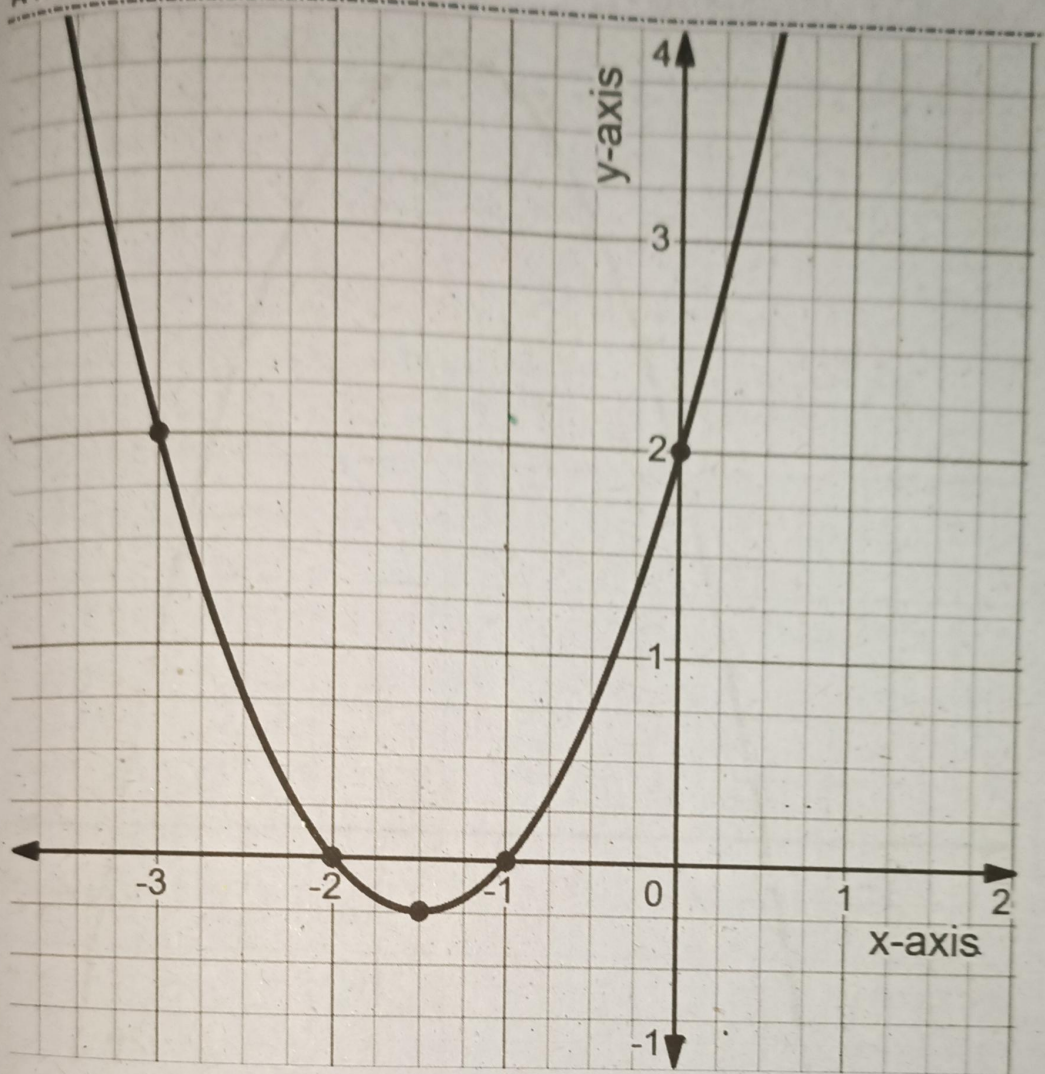
(ii) $y = -x^2 - 5x$

SOLUTION:

(i) $y = x^2 + 3x + 2$

$$y = (x)^2 + 2(x) \left(\frac{3}{2}\right) + \left(\frac{3}{2}\right)^2 - \left(\frac{3}{2}\right)^2 + 2$$

$$y = \left(x + \frac{3}{2}\right)^2 - \frac{1}{4}$$



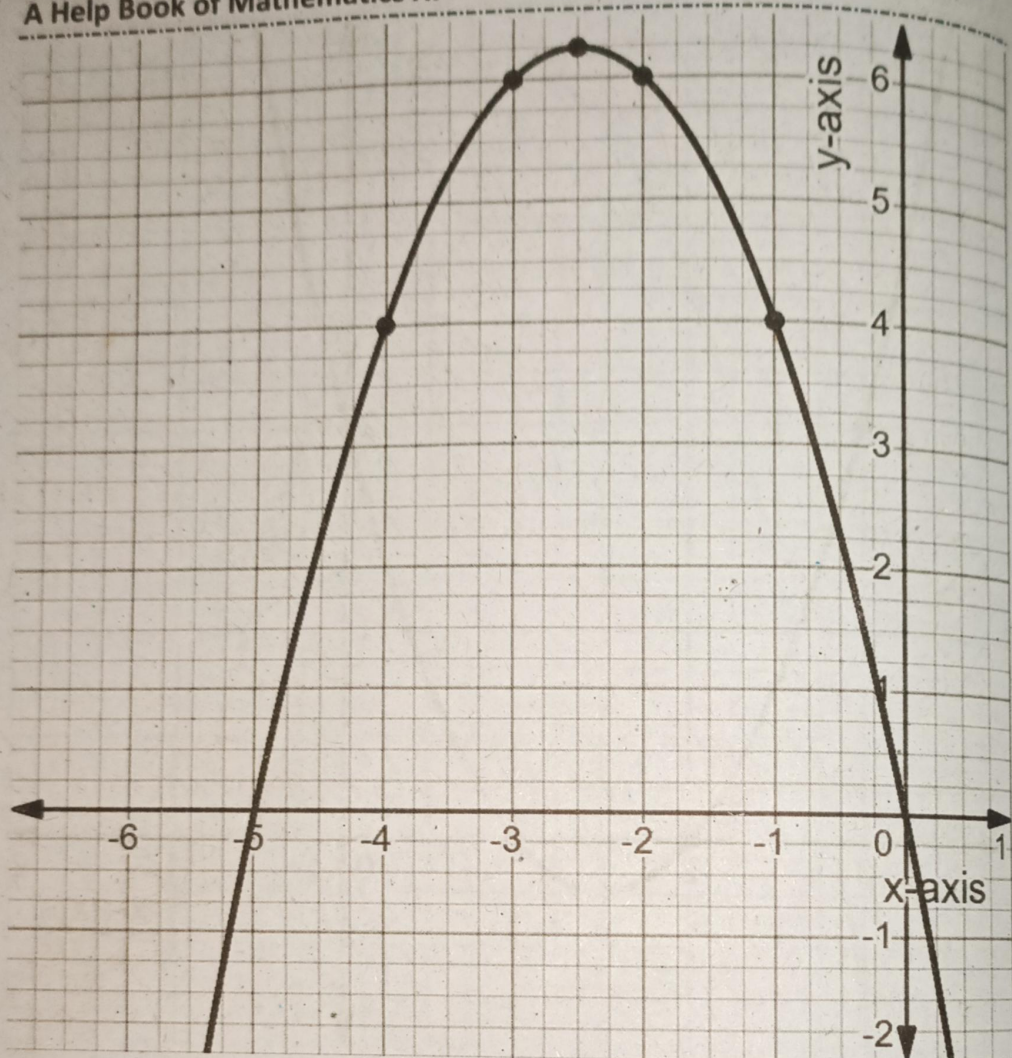
$$(ii) y = -x^2 - 5x$$

$$y = -(x^2 + 5x)$$

$$y = -\left\{(x)^2 + 5(x)\left(\frac{5}{2}\right) + \left(\frac{5}{2}\right)^2 - \left(\frac{5}{2}\right)^2\right\}$$

$$y = -\left\{\left(x + \frac{5}{2}\right)^2 - \frac{25}{4}\right\}$$

$$y = \frac{25}{4} - \left(x + \frac{5}{2}\right)^2$$



Q.5 Find the equation of the function of the type $y = f(x) = ax^2 + bx + c$ which cuts the x -axis at the points $(-4, 0)$ and $(3, 0)$ also passes through the point $(2, -4)$.

SOLUTION:

$$y = f(x) = ax^2 + bx + c$$

$$(-4, 0), (3, 0) \text{ and } (2, -4)$$

$$f(-4) = a(-4)^2 + b(-4) + c$$

$$0 = 16a - 4b + c$$

$$c = 4b - 16a \rightarrow (1)$$

$$f(3) = a(3)^2 + b(3) + c$$

$$0 = 9a + 3b + (4b - 16a)$$

$$-7a + 7b = 0$$

$$7b = 7a \Rightarrow \boxed{a = b} \rightarrow (2)$$

$$f(2) = a(2)^2 + b(2) + c$$

$$-4 = 4a + 2b + (4b - 16a)$$

$$-4 = -12a + 6b$$

$$-4 = -12a + 6a$$

$$-4 = -6a$$

$$\boxed{a = \frac{2}{3}}$$

$$(2) \Rightarrow \boxed{b = \frac{2}{3}}$$

$$(1) \Rightarrow c = 4\left(\frac{2}{3}\right) - 16\left(\frac{2}{3}\right)$$

$$\boxed{c = -8}$$

$$y = \frac{2}{3}x^2 + \frac{2}{3}x - 8$$

$$y = \frac{2x^2 + 2x - 24}{3}$$

$$\boxed{y = \frac{2(x^2 + x - 12)}{3}}$$

Q.6 Find the equation of the function of the type $y = f(x) = ax^2 + bx + c$ which crosses the x -axis at the points $(-8, 0)$ and $(9, 0)$ also passes through the point $(5, 10)$.

SOLUTION:

$$y = f(x) = ax^2 + bx + c$$

$(-8, 0), (9, 0)$ and $(5, 10)$

A Help Book of Mathematics XI

$$f(-8) = a(-8)^2 + b(-8) + c$$

$$0 = 64a - 8b + c$$

$$c = 8b - 64a \rightarrow (1)$$

$$f(9) = a(9)^2 + b(9) + c$$

$$0 = 81a + 9b + (8b - 64a)$$

$$17a + 17b = 0$$

$$17a = -17b$$

$$a = -b \rightarrow (2)$$

$$f(5) = a(5)^2 + b(5) + c$$

$$10 = 25a + 5b + (8b - 64a) \rightarrow (3)$$

$$10 = 39b + 13b$$

$$52b = 10$$

$$b = \frac{10}{52}$$

$$b = \frac{5}{26}$$

$$(2) \Rightarrow a = -\frac{5}{26}$$

$$(1) \Rightarrow c = 8\left(\frac{5}{26}\right) - 64\left(-\frac{5}{26}\right)$$

$$c = \frac{180}{26}$$

$$y = -\frac{5}{26}x^2 + \frac{5}{26}x + \frac{180}{26}$$

OR

$$f(x) = a(x+8)(x-9)$$

It passes through (5,10)

$$f(5) = a(5 + 8)(5 - 9)$$

$$10 = a(13)(-4)$$

$$a = -\frac{10}{(13)(-4)}$$

$$\boxed{a = \frac{5}{26}}$$

$$f(x) = \frac{5}{26}(x + 8)(x - 9)$$

$$f(x) = \frac{5}{26}(x^2 - x - 72)$$

Q.7 Find the equation of the function of the type $y = 3x^2 + bx + c$ of the parabola which cuts x -axis at the points $(-3, 0)$ and $(5, 0)$.

SOLUTION:

$(-3, 0)$ and $(5, 0)$

$$f(x) = 3x^2 + bx + c$$

$$f(-3) = 3(-3)^2 + b(-3) + c$$

$$0 = 27 - 3b + c$$

$$c = 3b - 27 \rightarrow (1)$$

$$f(5) = 3(5)^2 + b(5) + c$$

$$0 = 75 + 5b + (3b - 27)$$

$$8b = -48$$

$$\boxed{b = -6}$$

$$(1) \Rightarrow c = 3(-6) - 27$$

$$\boxed{c = -45}$$

$$f(x) = 3x^2 - 6x - 45$$

Q.8 Find the equation of the function of the type $y = 2x^2 + bx + c$ of the parabola which cuts x -axis at the points $(-4, 0)$ and $(5, 0)$.

SOLUTION:

$$f(x) = 2x^2 + bx + c$$

$$(-4, 0) \text{ and } (5, 0)$$

$$f(x) = 2x^2 + bx + c$$

$$f(-4) = 2(-4)^2 + b(-4) + c$$

$$0 = 32 - 4b + c$$

$$c = 4b - 32 \rightarrow (1)$$

$$f(5) = 2(5)^2 + b(5) + c$$

$$0 = 50 + 5b + (4b - 32)$$

$$9b = -18$$

$$\boxed{b = -2}$$

$$(1) \Rightarrow c = 4(-2) - 32$$

$$\boxed{c = -40}$$

$$f(x) = 2x^2 - 2x - 40$$

EXERCISE 8.4

Q.1 Graphically find the points of intersection of the following linear functions with co-ordinate axes.

(i) $f(x) = x + 7$

(ii) $f(x) = -x + 5$

(iii) $f(x) = 3x + 2$

(iv) $f(x) = -2x + 4$

SOLUTION:

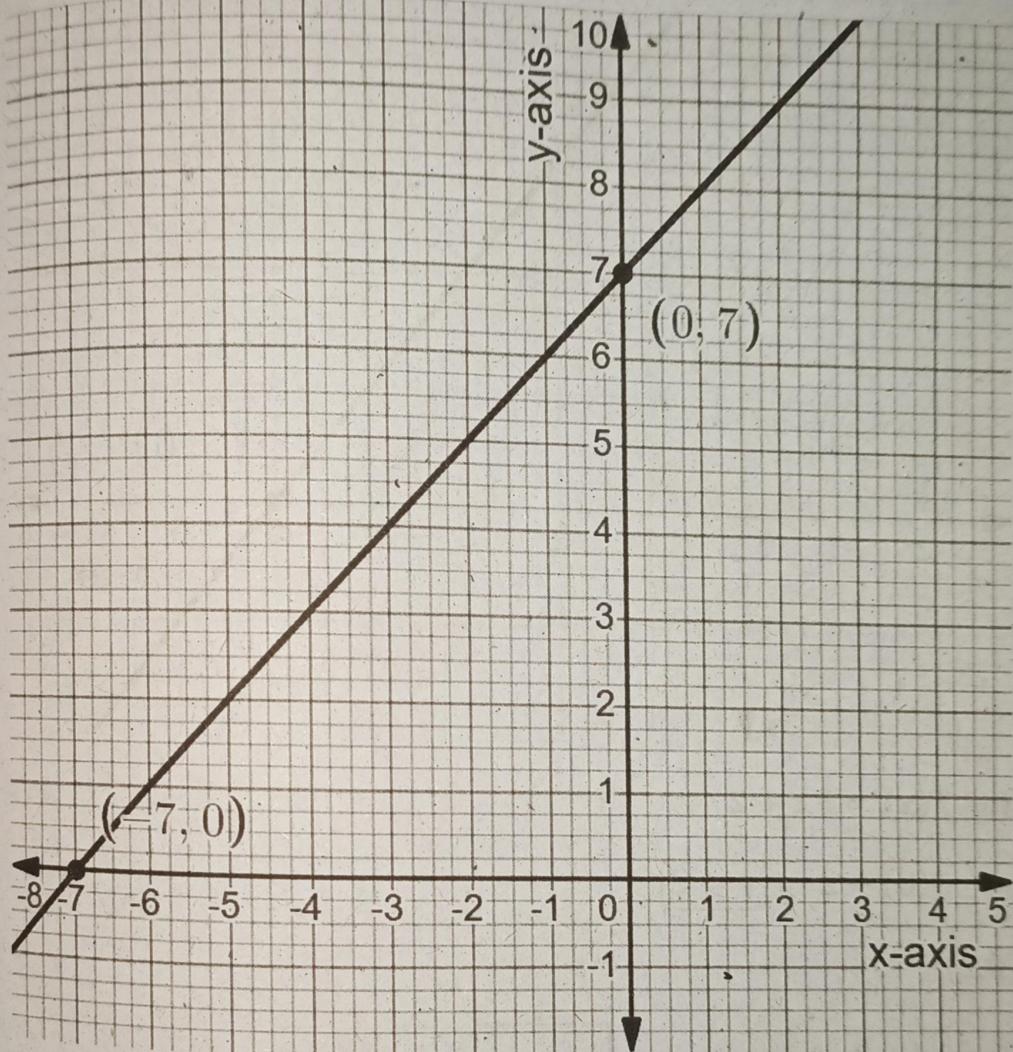
(i) $f(x) = x + 7$

$$y = 0 + 7 \Rightarrow \boxed{y = 7}$$

$(0, 7)$

$$0 = x + 7 \Rightarrow \boxed{x = -7}$$

$(-7, 0)$



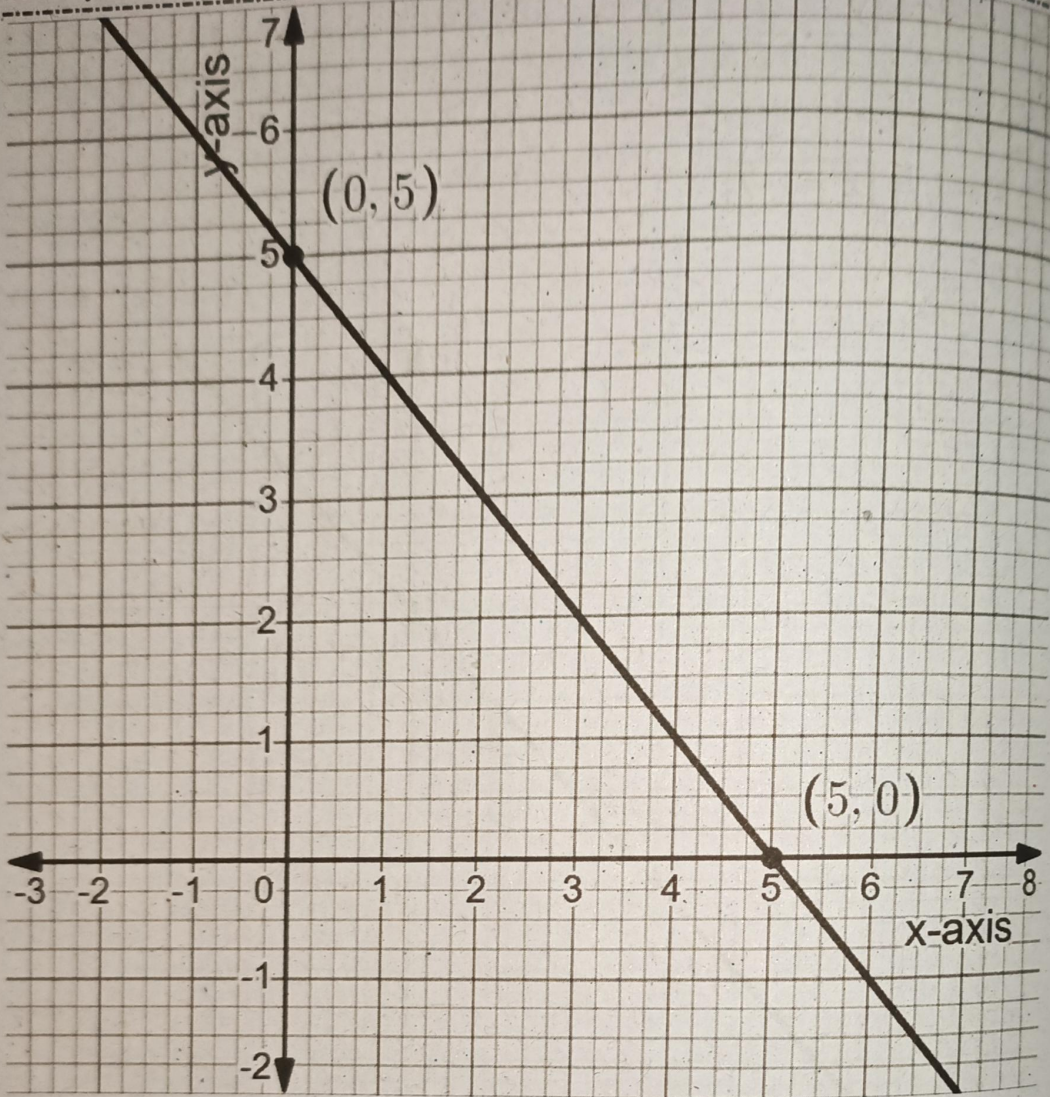
(ii) $f(x) = -x + 5$

$$y = 0 + 5 \Rightarrow \boxed{y = 5}$$

$(0, 5)$

$$0 = -x + 5 \Rightarrow \boxed{x = 5}$$

$(5, 0)$



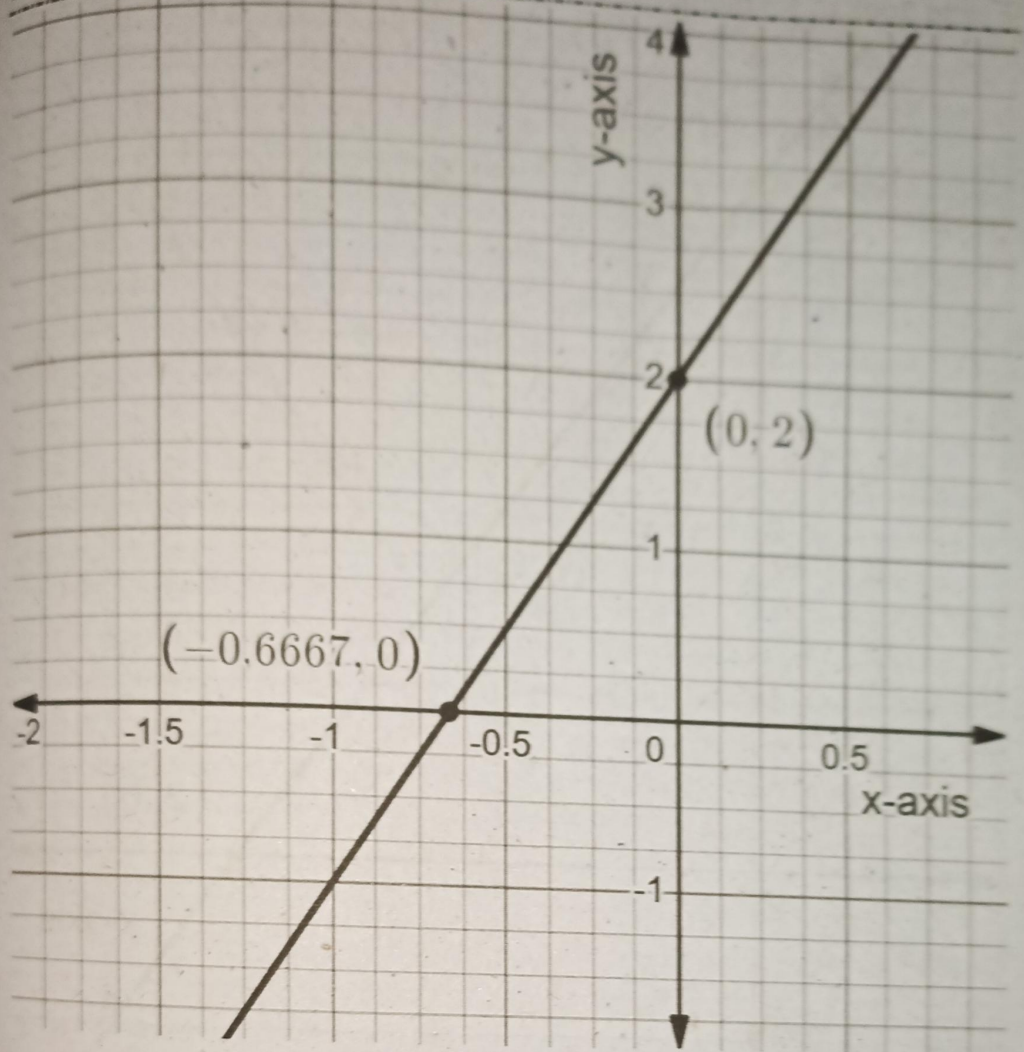
$$(iii) f(x) = 3x + 2$$

$$y = 3(0) + 2 \Rightarrow \boxed{y = 2}$$

$$(0, 2)$$

$$0 = 3x + 2 \Rightarrow 3x = -2 \Rightarrow \boxed{x = -\frac{2}{3}}$$

$$\left(-\frac{2}{3}, 0\right)$$



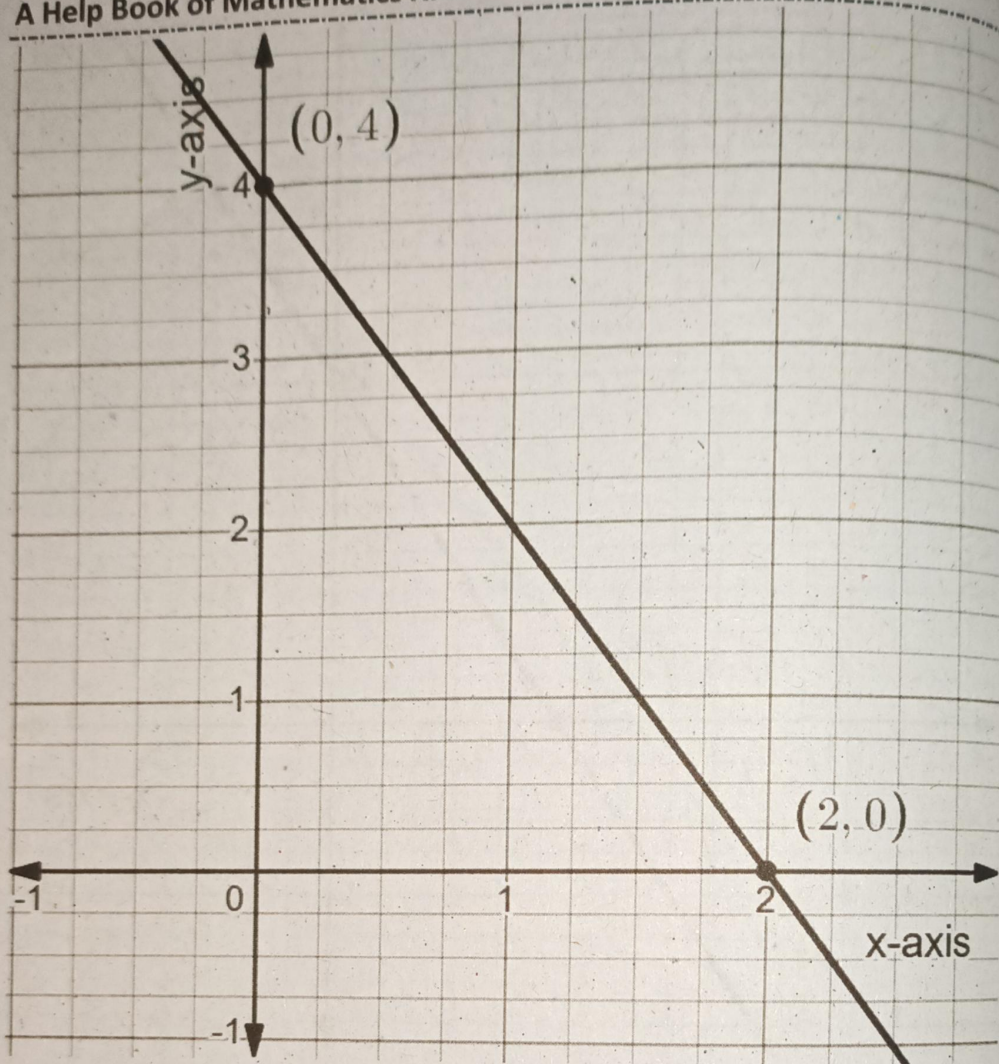
$$(iv) f(x) = -2x + 4$$

$$y = -2(0) + 4 \Rightarrow \boxed{y = 4}$$

$$(0, 4)$$

$$0 = -2x + 4 \Rightarrow 2x = 4 \Rightarrow \boxed{x = 2}$$

$$(2, 0)$$



Q.2 Find the point of intersection of the following functions graphically where y is function of x .

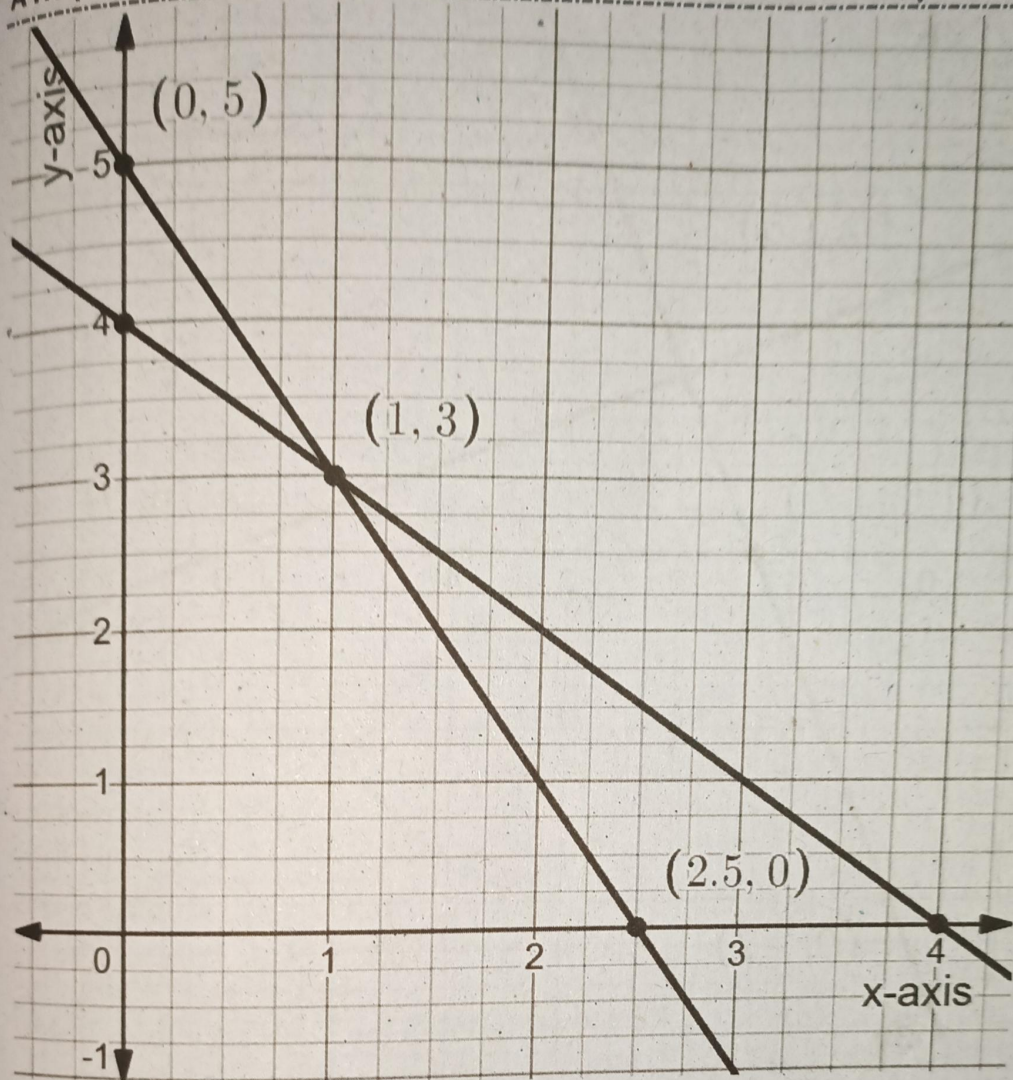
- (i) $2x + y = 5$ and $x + y = 4$ (ii) $3x - 2y = 4$ and $x + 4y = 6$

SOLUTION:

- (i) $2x + y = 5$ and $x + y = 4$

x	0	$\frac{5}{2}$	1
$y = 5 - 2x$	5	0	3

x	0	4	1
$y = 4 - x$	4	0	3



(ii) $3x - 2y = 4$ and $x + 4y = 6$

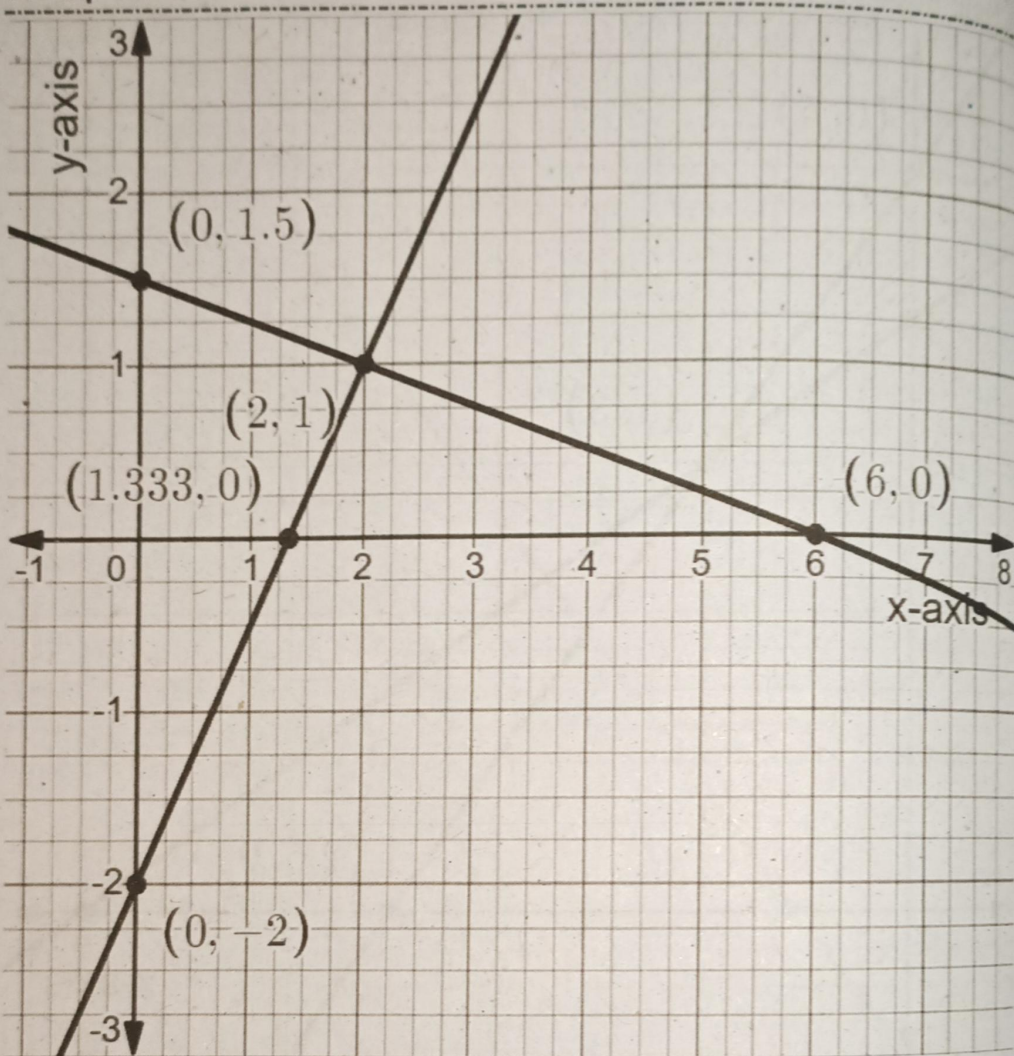
$$3x - 2y = 4$$

$$3x - 4 = 2y$$

x	0	$\frac{4}{3}$	2
$y = \frac{3x-4}{2}$	-2	0	1

$$x + 4y = 6 \Rightarrow 4y = 6 - x \Rightarrow y = \frac{6-x}{4}$$

x	6	0	2
$y = \frac{3x-4}{2}$	0	$\frac{3}{2}$	1



Q.3 Find the point of intersection of the following functions graphically:

(i) $f(x) = x + 2$ and $g(x) = x^2 - 4x + 6$

(ii) $f(x) = x + 4$ and $g(x) = x^2 - 6x + 10$

SOLUTION:

(i) $f(x) = x + 2$ and $g(x) = x^2 - 4x + 6$

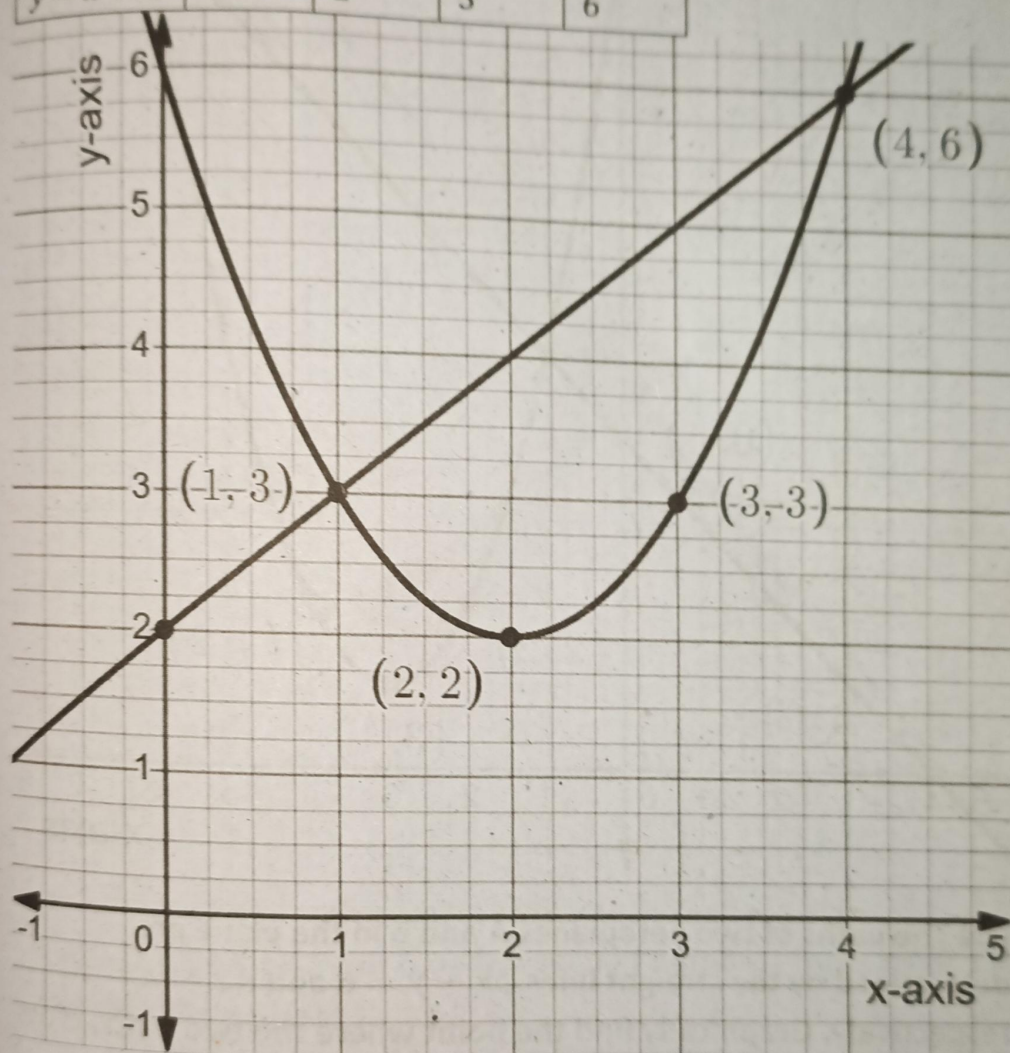
x	-2	0	1	4
$y = x + 2$	0	2	3	6

$$g(x) = x^2 - 4x + 6$$

$$g(x) = x^2 - 4x + 4 + 2$$

$$g(x) = (x - 2)^2 + 2$$

x	1	2	3	4
$y = x + 2$	3	2	3	6



(ii) $f(x) = x + 4$ and $g(x) = x^2 - 6x + 10$

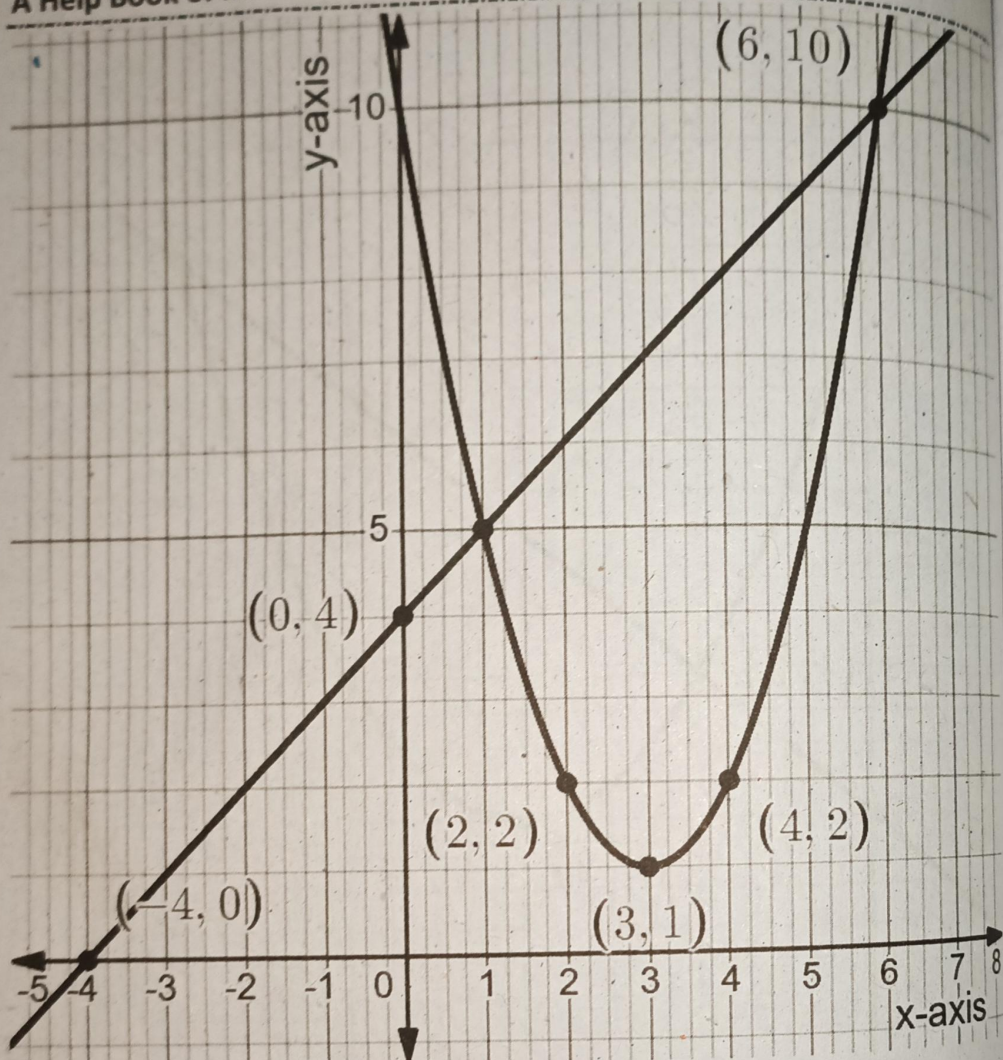
x	-4	0	1	4
$y = x + 4$	0	4	3	6

$$g(x) = x^2 - 4x + 6$$

$$g(x) = x^2 - 4x + 4 + 2$$

$$g(x) = (x - 2)^2 + 2$$

x	1	2	3	4
$y = x + 2$	3	2	3	6



Q.4 The paths of two aeroplanes A and B in the plane are determined by the straight lines $2x - y = 6$ and $3x + y = 4$ respectively. Graphically find the point where the two paths cross each other.

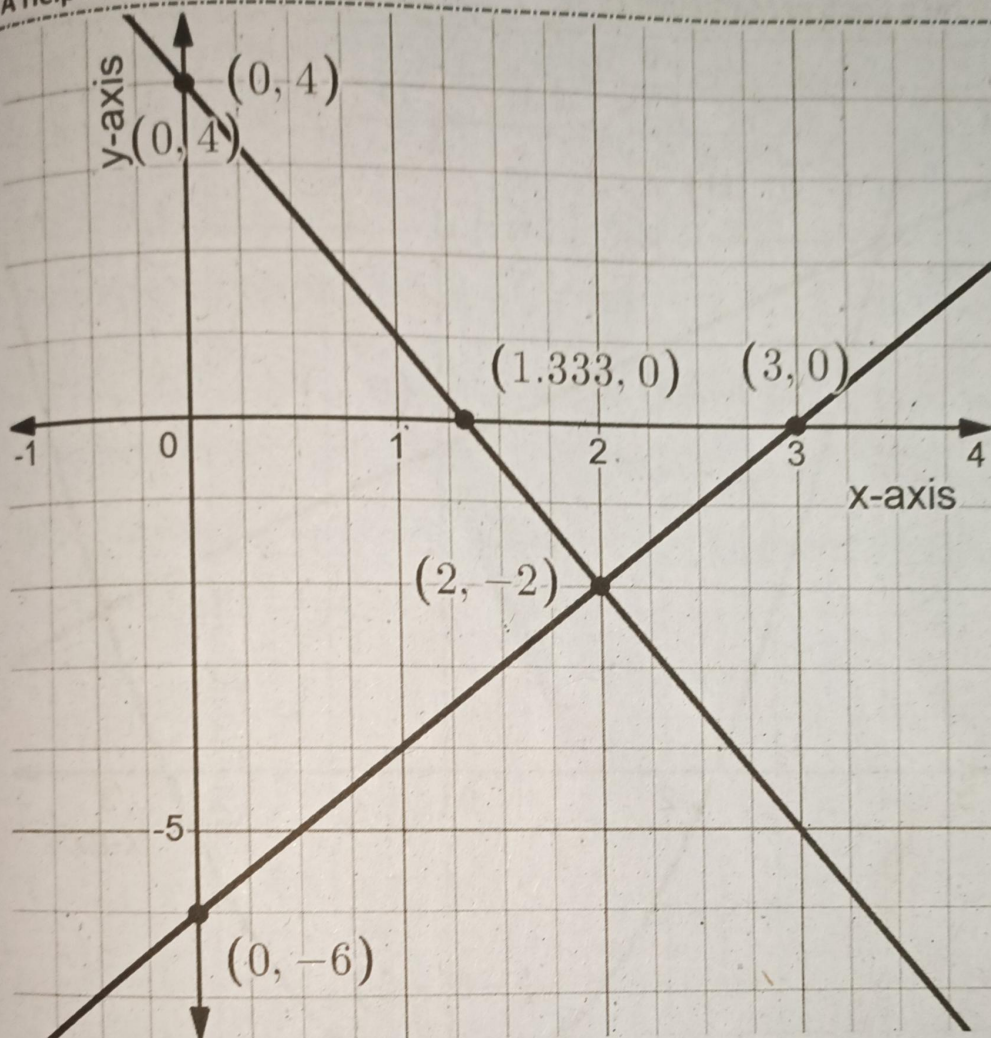
SOLUTION:

$$2x - y = 6$$

x	0	3	2
$y = 2x - 6$	-6	0	-2

$$3x + y = 4 \Rightarrow y = 4 - 3x$$

x	$\frac{4}{3}$	0	2
$y = 4 - 3x$	0	4	-2



Q.5 If the sum of two number is 6 and the square of the first number is greater than 6 by the second number. Solve graphically and find the numbers.

SOLUTION:

Let the numbers are x and y

$$x + y = 6$$

$$y = 6 - x$$

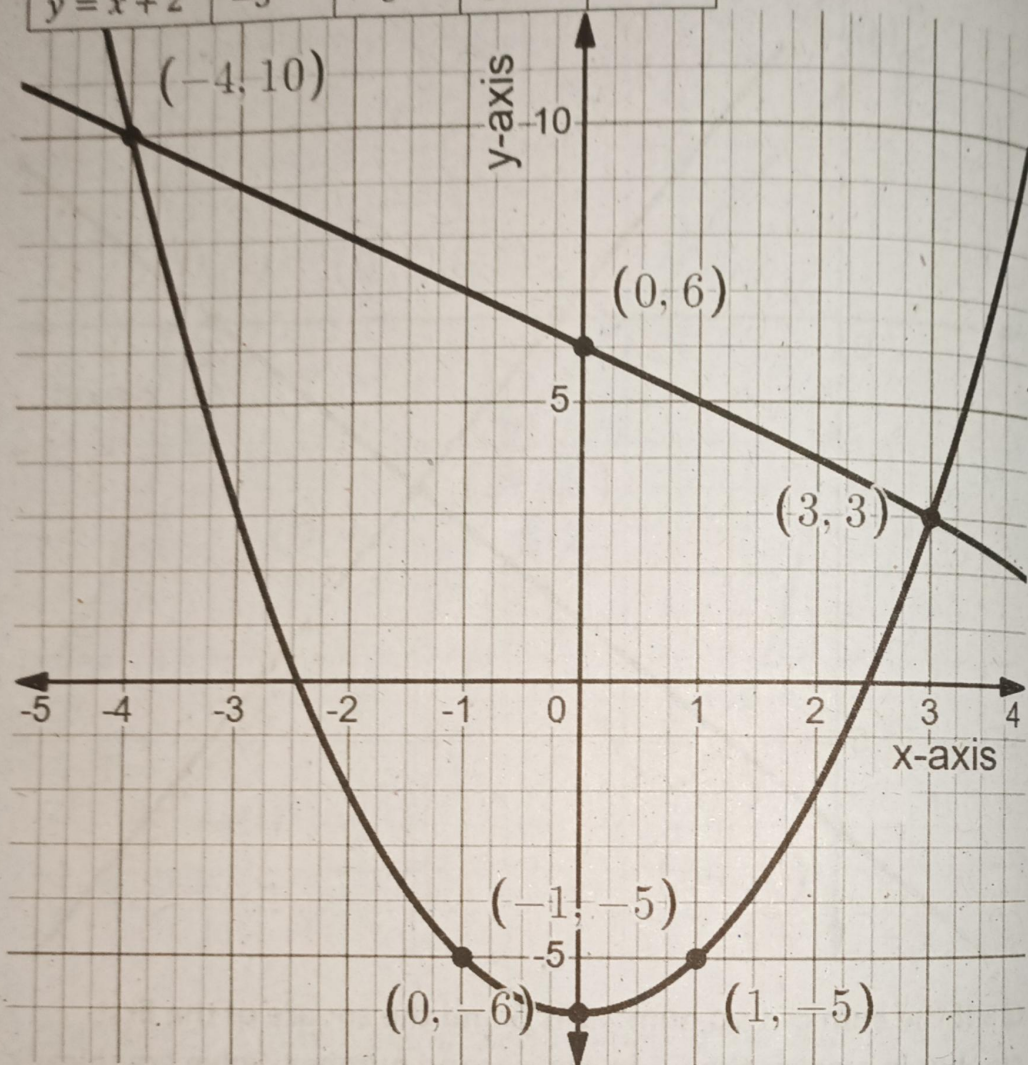
$$x^2 = y + 6$$

$$y = x^2 - 6$$

x	-6	0	3	-4
$y = 6 - x$	0	6	3	10

$$g(x) = x^2 - 6$$

x	-1	0	3	-4
$y = x + 2$	-5	-6	3	10



REVIEW EXERCISE 8

1. Multiple Choice Question.

(i) If $f: A \rightarrow B$ be a function, then it is an onto function, if:

(a) $\text{Range} = B$

(b) $\text{Range} \subset B$

(c) Image is not repeated

(d) $\text{Domain} \neq A$

(ii) The graph of $y = x^6$ is symmetric to _____.

(a) x -axis

(b) y -axis

(c) origin

(d) none

(iii) An one to one function is also called _____ function.

- (a) injective
- (b) surjective
- (c) bijective
- (d) inverse

(iv) Inverse of a function exists only if it is:

- (a) injective
- (b) bijective
- (c) surjective
- (d) all of these

(v) The function $f = \{(x, y) | y = mx + c, a \neq 0\}$, m and c are real numbers is:

- (a) linear
- (b) quadratic
- (c) a circle
- (d) a point

(vi) The function $f = \{(x, y) | y = ax^2 + bx + c, a \neq 0\}$ is:

- (a) linear
- (b) quadratic
- (c) a circle
- (d) a point

(vii) The range of $f(x) = x^2 + 3, \forall x \in \mathbb{R}$ is _____.

- (a) $\mathbb{R}$
- (b) $\mathbb{R}^{-1}$
- (c) $(3, \infty)$
- (d) $[3, \infty]$

(viii) Graph of $y = x^n$ is symmetric to _____ if n is an odd integer.

- (a) x -axis
- (b) y -axis
- (c) origin
- (d) none

(ix) Graph of linear function is:

- (a) circle
- (b) straight line
- (c) parabola
- (d) triangle

(x) If f is function from A to B, then domain of f is equal to:

- (a) any subset of A
(b) $A \times B$
(c) A
(d) B

(xi) Every function is a:

- (a) relation
(b) inverse function
(c) one to one
(d) none of these

(xii) If f and g are equal where $f(x) = 7x - 4$ and $g(x) = x$ then $x =$:

- (a) $\frac{1}{3}$
(b) $\frac{2}{3}$
(c) 1
(d) $\frac{4}{3}$

(xiii) Domain of the function $f(x) = \frac{3x+1}{x-1}$ is set of all:

- (a) real numbers
(b) rational numbers
(c) complex numbers
(d) real numbers except 1

(xiv) A function $f(x) = |x| - x^2$ is:

- (a) odd
(b) linear
(c) even
(d) neither even nor odd

Q.2 Find the domain of the following functions:

(i) $f(x) = \frac{2x-1}{x+4}$

(ii) $f(x) = \sqrt{x+3}$

SOLUTION:

(i) $f(x) = \frac{2x-1}{x+4}$

$x + 4 \neq 0$

$x \neq -4$

Domain = $\mathbb{R} - \{-4\}$

(ii) $f(x) = \sqrt{x+3}$

$x+3 \geq 0$

$x \geq -3$

Domain = $[-3, \infty)$

Q.3 Graphically, find the point of intersection of the function

$f(x) = x$ and $g(x) = x^2 - x$.

$y = x$

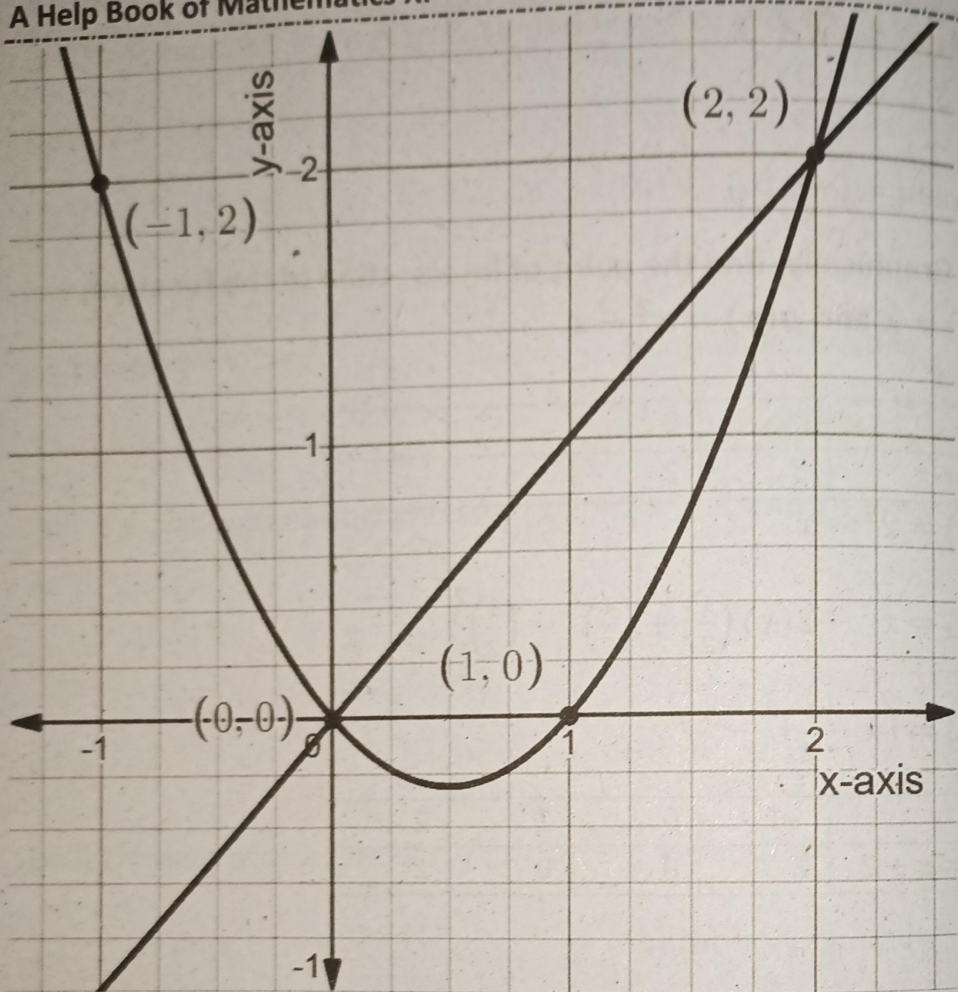
x	0	2		
$y = x$	0	2		

$g(x) = x^2 - x$

$g(x) = x^2 - 2(x) \left(\frac{1}{2}\right) + \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2$

$g(x) = \left(x - \frac{1}{2}\right)^2 - \frac{1}{4}$

x	-1	0	1	2
$y = x + 2$	2	0	0	2



Q.4 Find the equation, in the form $y = ax^2 + bx + c$ of the parabola which cuts x -axis at (1, 0) and (5, 0) and cuts y -axis at (0, 15).

Quadratic equation will be of the form

$$f(x) = a(x - 1)(x - 5)$$

It passes through

$$(0, 15)$$

$$15 = a(0 - 1)(0 - 5)$$

$$15 = a(-1)(-5)$$

$$5a = 15$$

$$\boxed{a = 3}$$

$$f(x) = 3(x - 1)(x - 5)$$

$$f(x) = 3(x^2 - 5x - x + 5)$$

$$f(x) = 3(x^2 - 6x + 5)$$

Q.5 The function f is defined by $f(x) = ax^2 + bx + c$. Given that $f(0) = 5$, $f(-1) = 15$ and $f(1) = 1$, find the values of a , b and c .

$$f(x) = ax^2 + bx + c$$

$$f(0) = a(0)^2 + b(0) + c$$

$$5 = c \Rightarrow \boxed{c = 5}$$

$$f(-1) = a(-1)^2 + b(-1) + 5$$

$$15 = a - b + 5$$

$$15 - 5 = a - b$$

$$a - b = 10 \rightarrow (1)$$

$$f(1) = a(1)^2 + b(1) + 5$$

$$1 = a + b + 5$$

$$a + b = 1 - 5$$

$$a + b = -4 \rightarrow (2)$$

$$(1) + (2)$$

$$2a = 6$$

$$\boxed{a = 3}$$

$$(2) \Rightarrow b = -3 - 4$$

$$\Rightarrow \boxed{b = -7}$$

$$f(x) = 3x^2 - 7x + 5$$

Q.6 The function f is defined by $f(x) = x^2 - 3x + 5$. Find the values of x for which $f(x) = 3$.

$$f(x) = x^2 - 3x + 5$$

$$3 = x^2 - 3x + 5$$

$$0 = x^2 - 3x + 5 - 3$$

$$x^2 - 3x + 2 = 0$$

$$x^2 - 2x - x + 2 = 0$$

$$x(x - 2) - (x - 2) = 0$$

$$(x - 2)(x - 1) = 0$$

$$\boxed{x = 1, 2}$$